TURTLE GAMES

Driving Customer Loyalty Through Data

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What Drives Customer Loyalty at Turtle Games?

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Key Project Objectives

1. Understand how customers **earn and engage** with **loyalty points**
2. Segment customers into groups to identify **targetable segments**
3. Leverage **customer review text** data to inform marketing and business decisions
4. Use **descriptive statistics** to assess loyalty points data for **predictive modelling**

Dataset & Tools Used

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Dataset

- **Content:** 2,000 entries from **turtle_reviews.csv**
- **Key variables:** Age, Gender, Remuneration, Spending Score, Education, Loyalty Points, Review, Summary

Tools



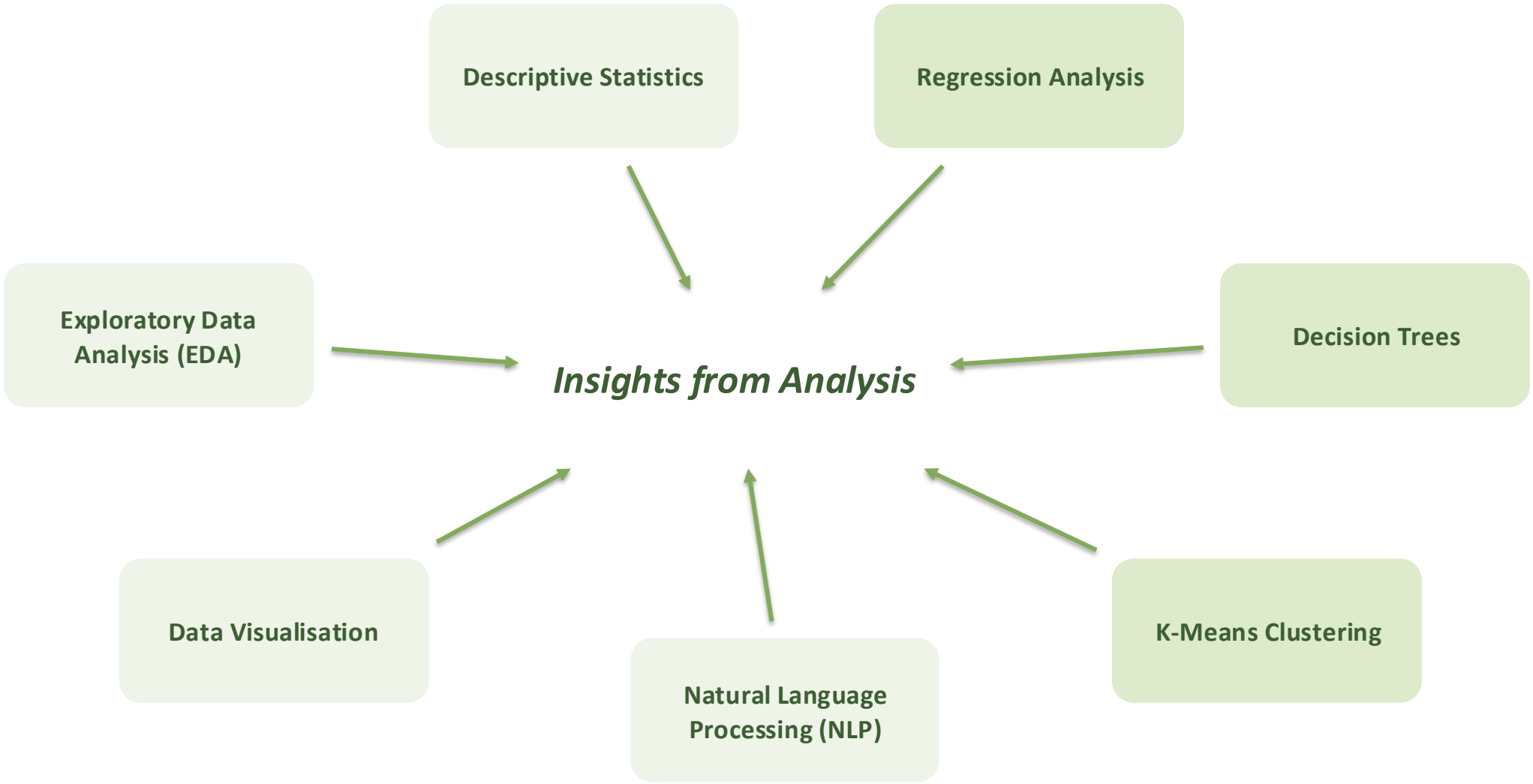
Python: Used for basic descriptive statistics, decision tree modelling, sentiment analysis



R: Exploratory Data Analysis, Multiple Linear Regression

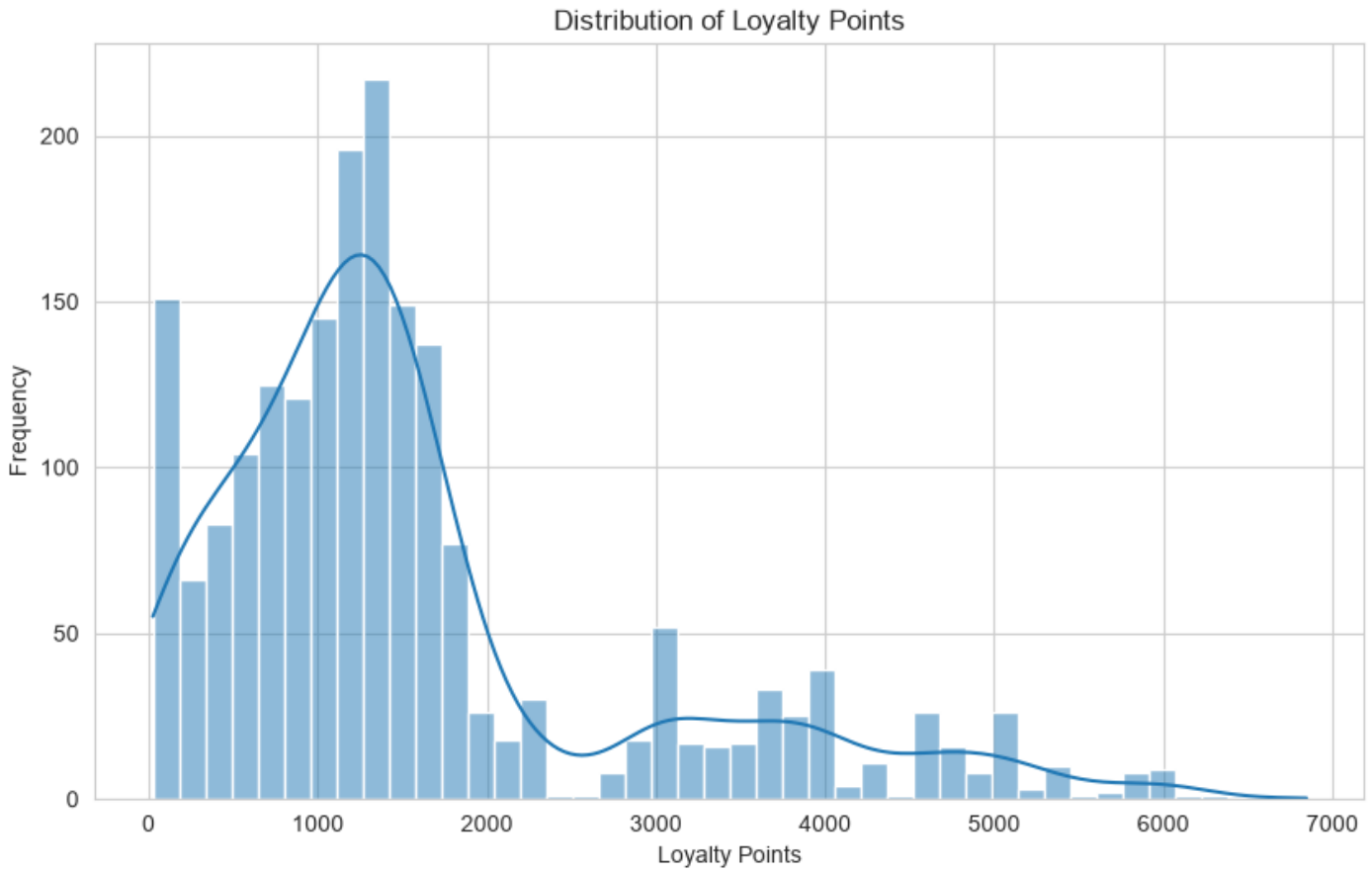
Analytical Approach

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Most Customers Hold Under 2,000 Loyalty Points

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Right-tailed distribution

1,578 Mean

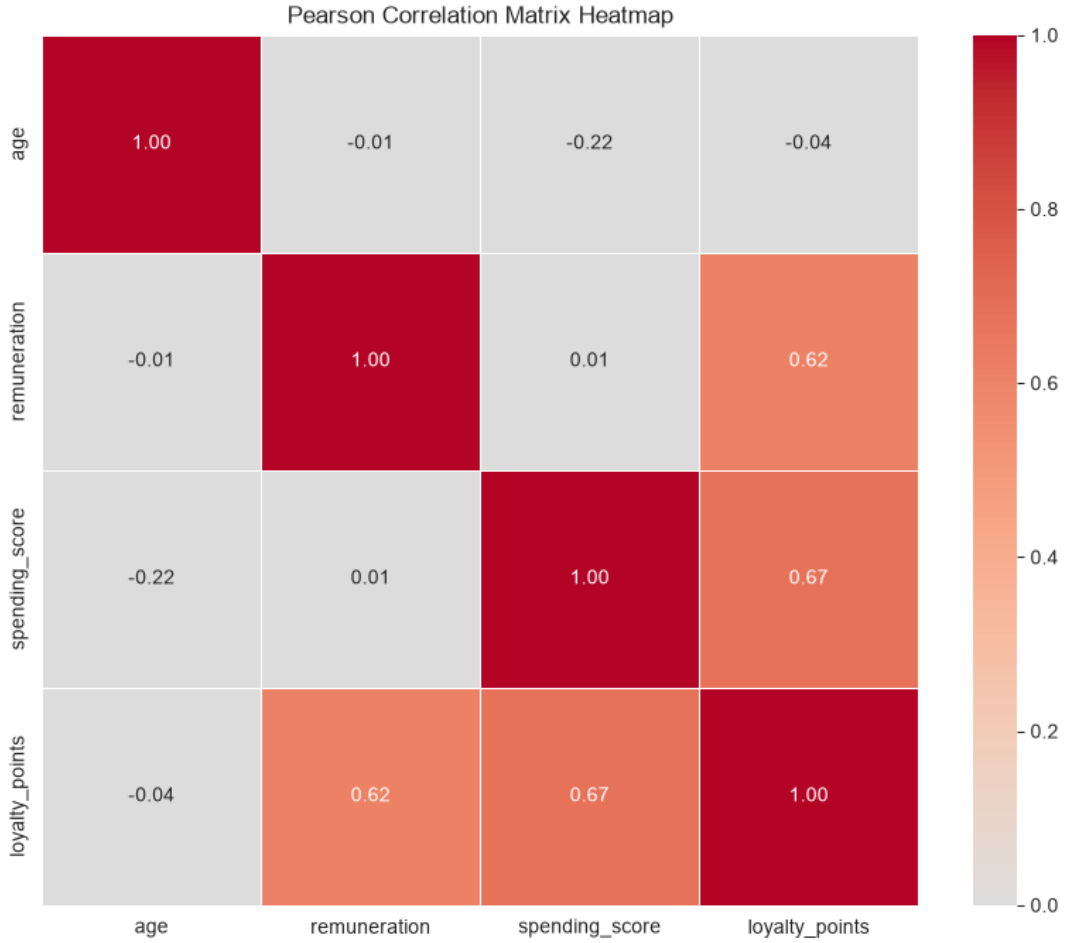
> **1,276** Median

The mean is higher than the median, showing that a **small number of customers with very high loyalty points** pulls the average upwards

Outliers were kept as they form a real segment of highly engaged customers, not data errors

Spending Score and Remuneration Drive Loyalty Points

Both correlate strongly with loyalty points but not with each other



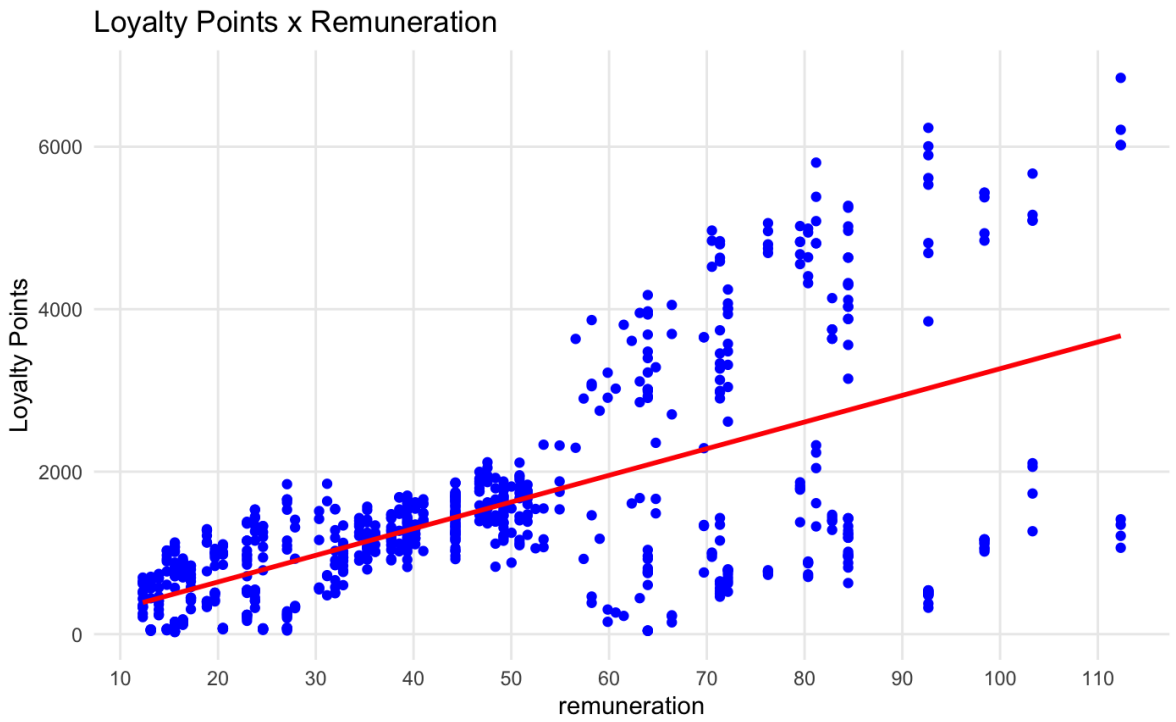
Pearson Correlation with Loyalty Points

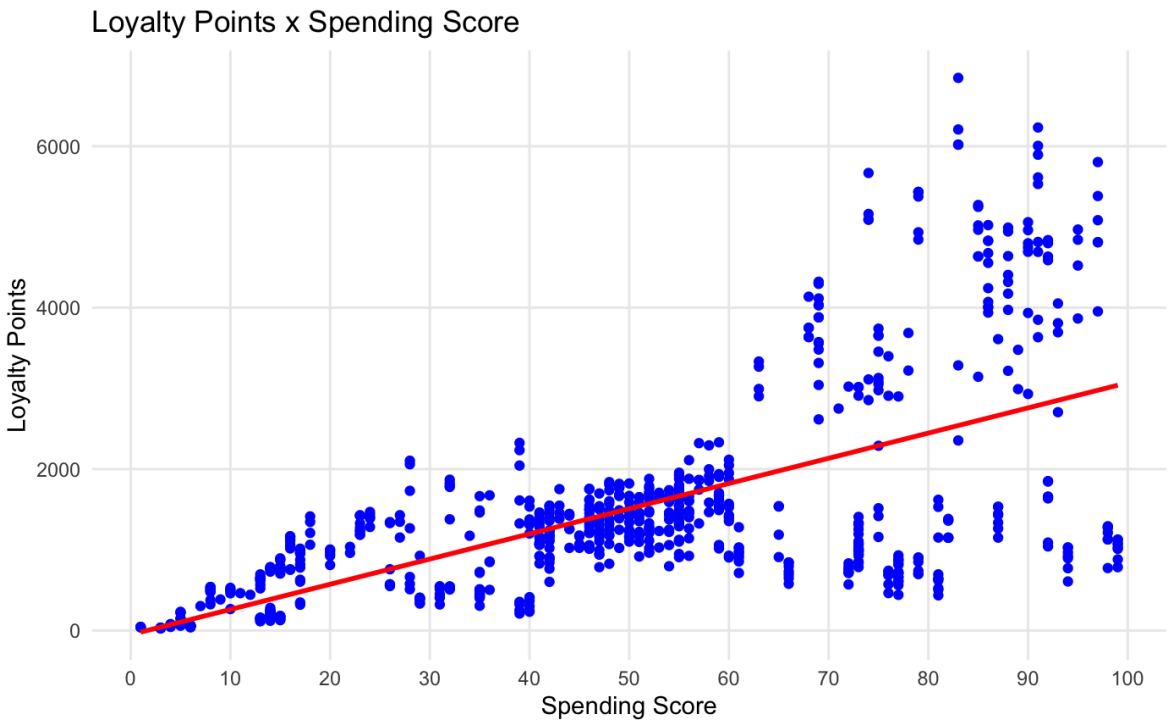
Spending score	**0.67**
Remuneration	**0.62**
Age	**-0.04**

Only **remuneration** and **spending score** have a positive relationship with loyalty points

High Points Need Both High Income and High Spending

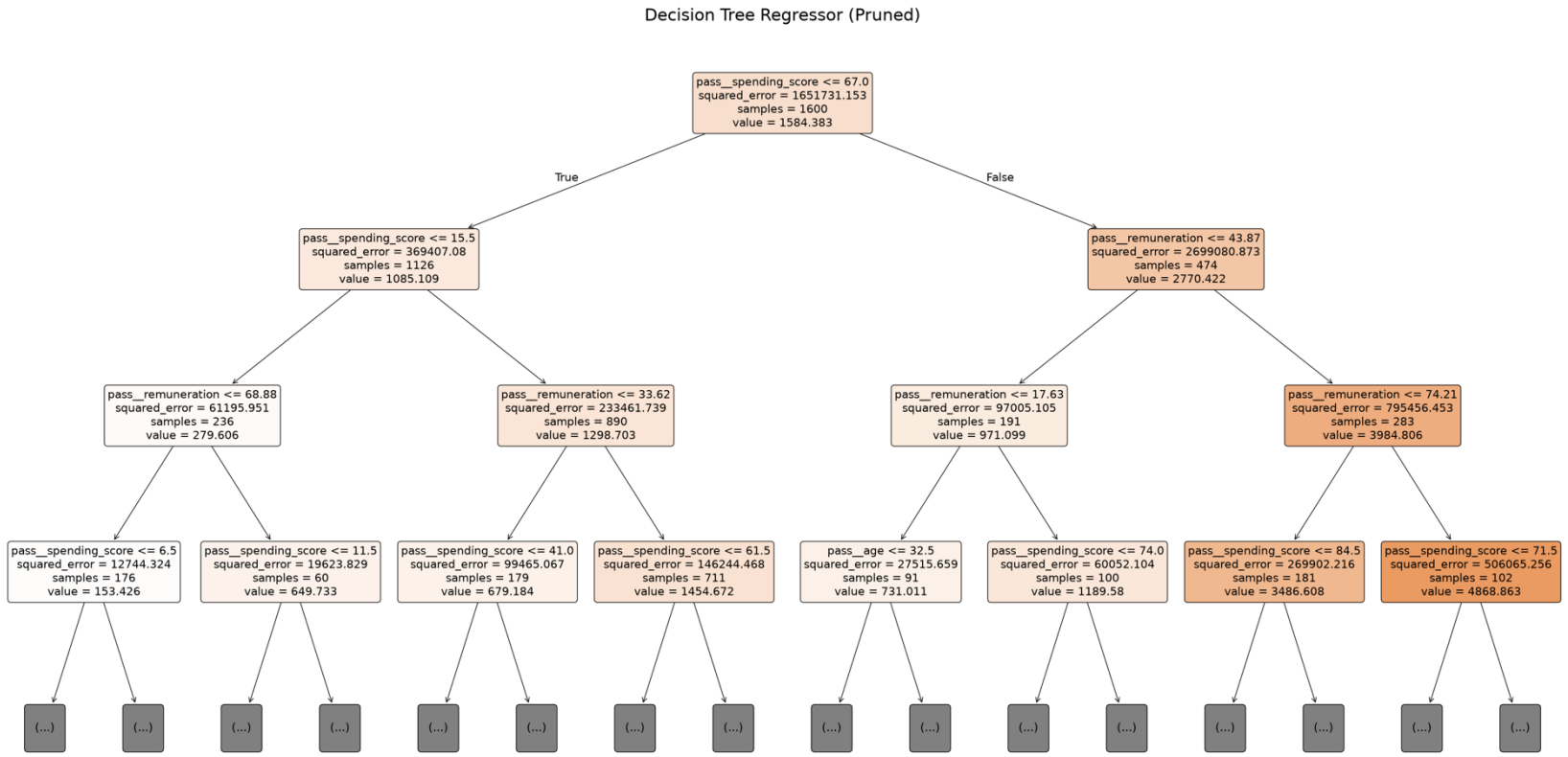
Each relationship splits into two groups at the top end





Visualising Loyalty Drivers with a Decision Tree

Spending score and remuneration consistently split customers into high and low loyalty earners

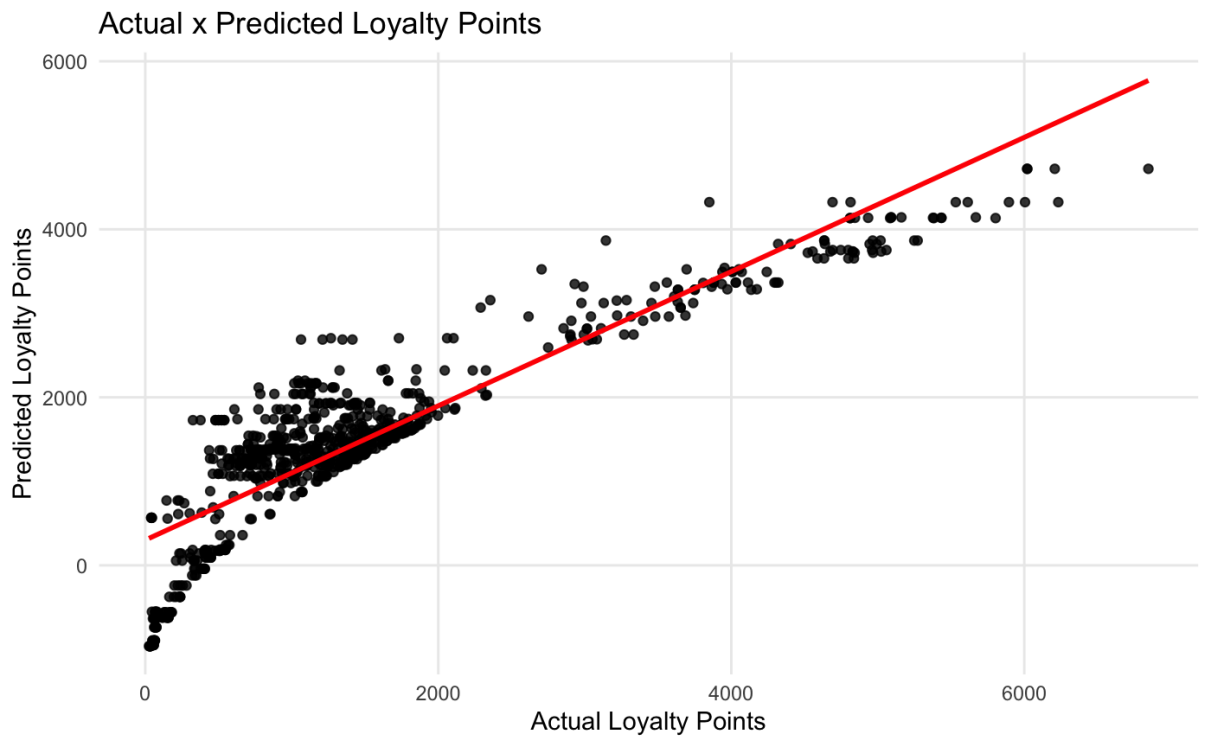


This pruned regression tree shows how customer traits influence loyalty points

Splits at the top of the tree reinforce the key predictors: **behaviour, not demographics, drives loyalty outcomes**

Predicting Loyalty Points Using Regression

Spending score and remuneration are statistically strong predictors of loyalty



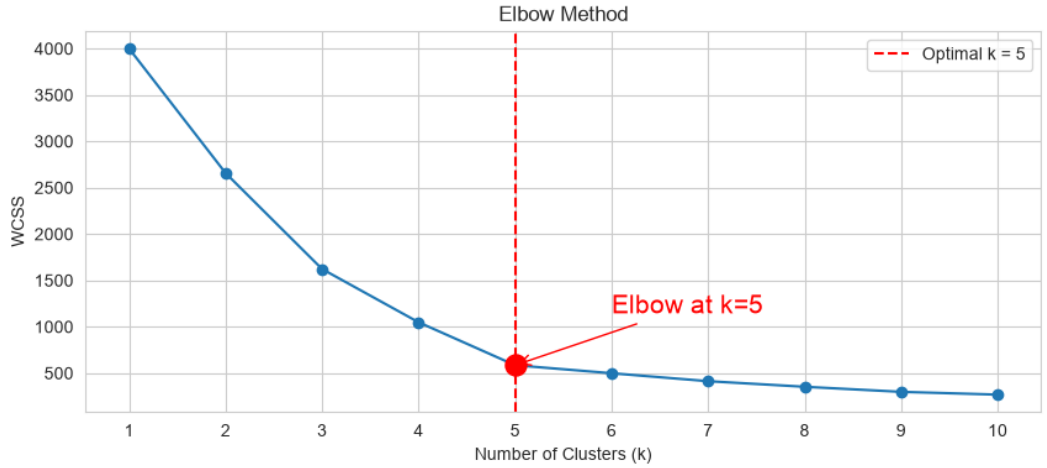
Regression Equation

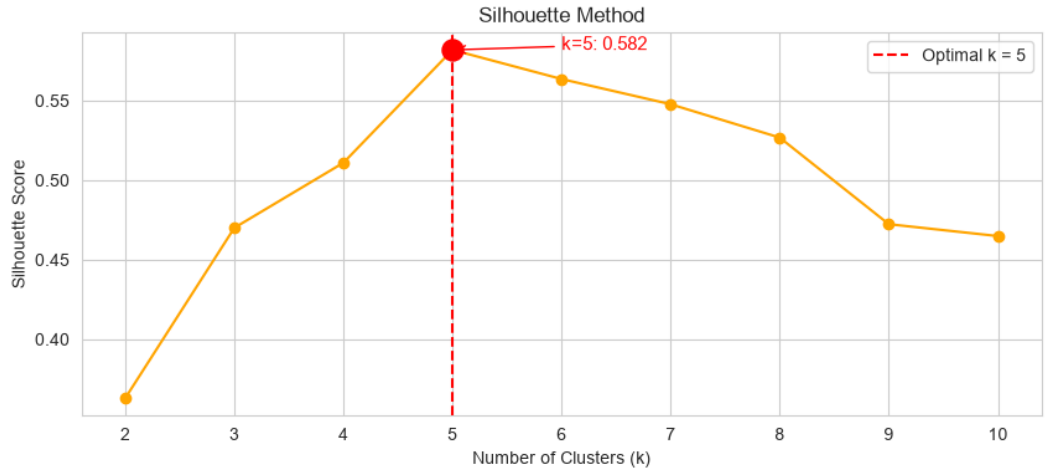
Loyalty Points = 33.98 × Remuneration + 32.89 × Spending Score − 1,700.31

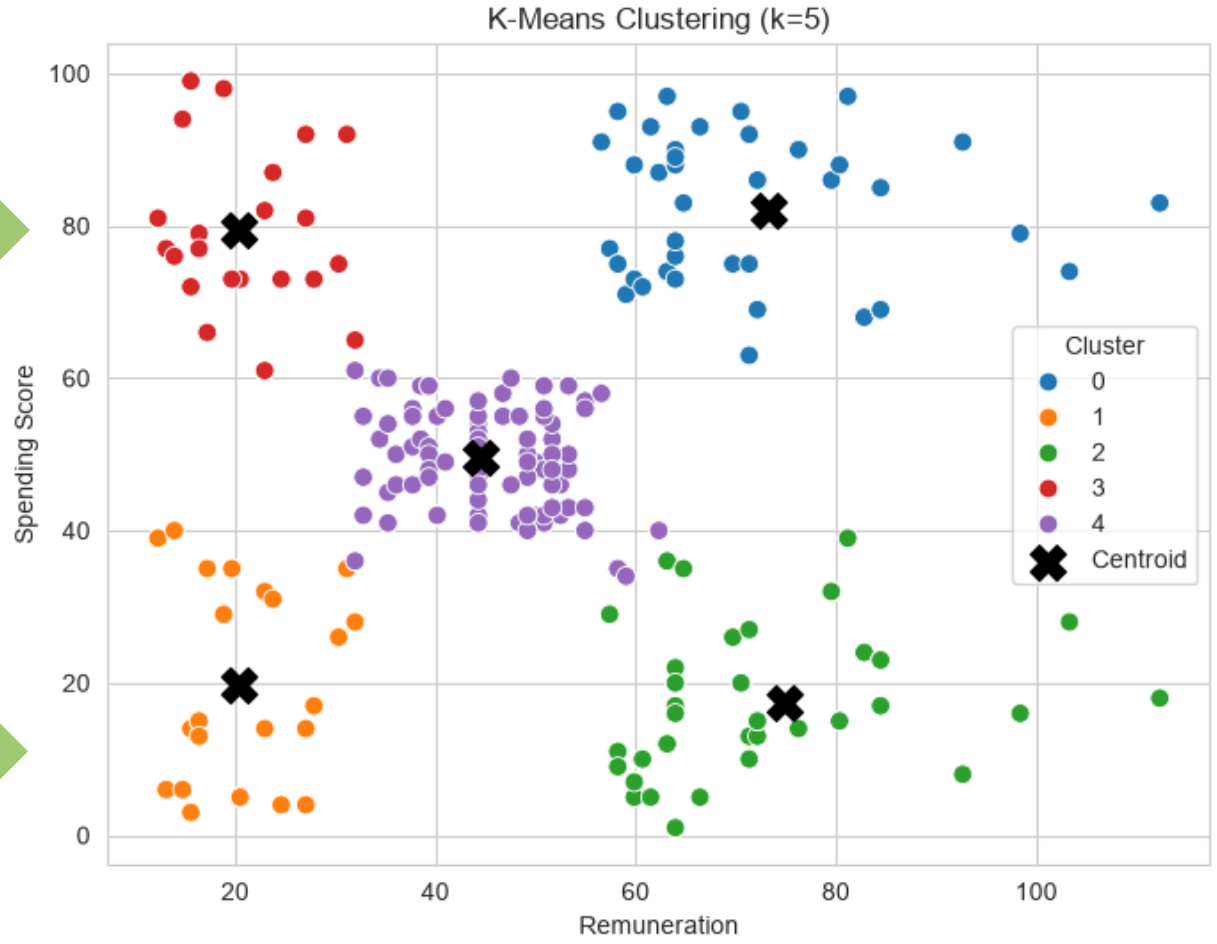
- **Spending score:** each one-point increase adds about 32.89 loyalty points, holding remuneration constant
- **Remuneration:** each one-unit (£1k) increase adds about 33.98 loyalty points, holding spending score constant
- **R^2 = 0.827:** a strong fit, with the two features explaining around 83% of the variation in loyalty points

Five Customer Segments by Income and Spending

Both Elbow method and Silhouette score point to k = 5

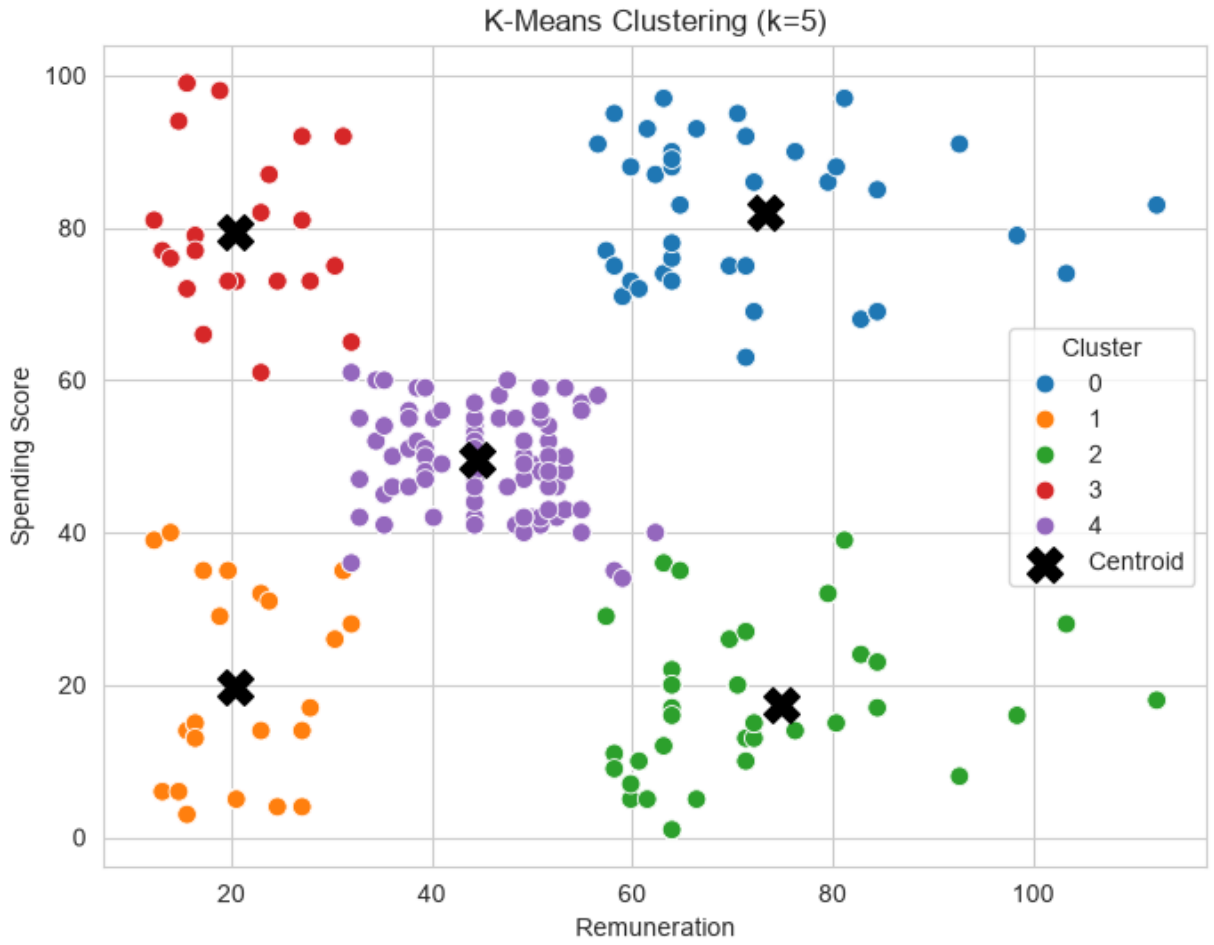






Each Segment Needs a Different Marketing Approach

High-income, high-spending customers hold about 20 times the points of the lowest segment



Segment	Profile	Recommended action
0	**High income, high spending**	**VIP programmes and personalized communication**
4	Average income and spending	Personalised offers
3	Low income, high spending	Loyalty rewards and exclusive offer to retain them
2	High income, low spending	Exclusive products and premium promotions to raise spending
1	Low income, low spending	Frequent and low-cost rewards

What Customers Talk About

Reviews describe products and how they are used; summaries are short ratings and praise

Reviews (Stopwords Removed)



Product lines stand out: cards, tiles, books, expansions. Kids, gifts and family appear often.

Summaries

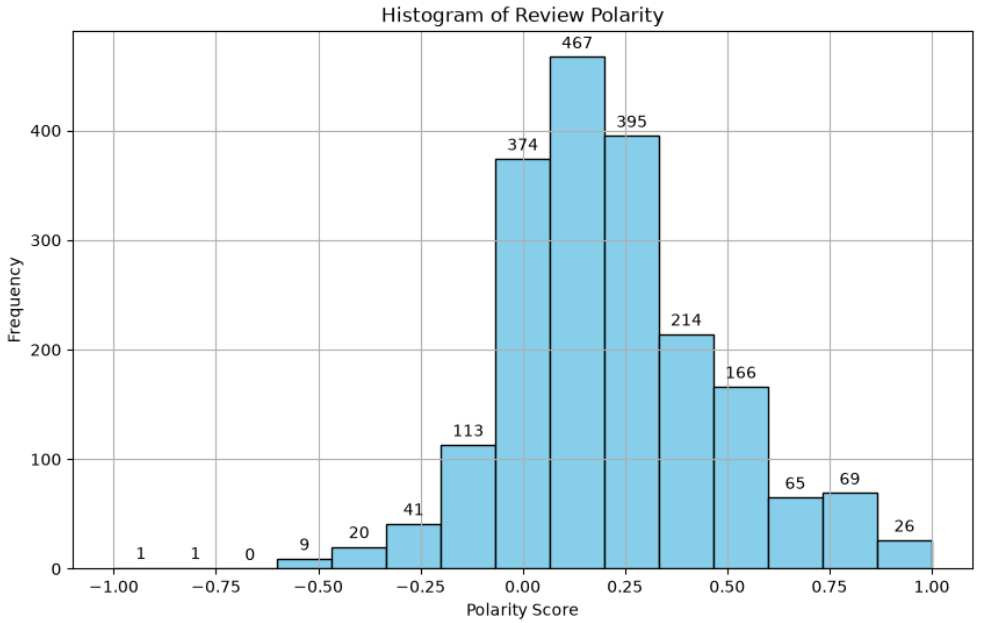


Dominated by 'five stars', 'great' and 'fun'. 378 summaries are only the default 'Five Stars'.

Review Sentiment Is Mostly Mildly Positive

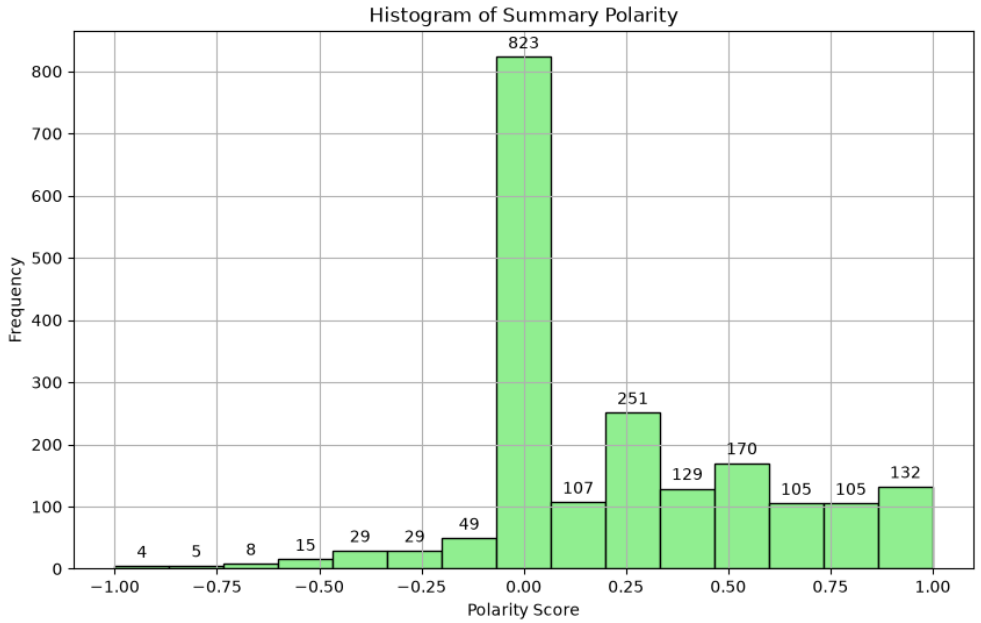
TextBlob polarity scores for full reviews and summaries

Reviews



Bell-shaped, centred slightly above 0. Most scores fall between 0 and 0.5.

Summaries



Spike at 0 is misleading: 'Five Stars' contains no sentiment words, so TextBlob scores it as neutral.

What Drives Positive and Negative Sentiment

Customers love gifting and learning; complaints are about usability and value

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Positive Reviews

1. **Gifts and learning**
   'Perfect for tutoring my grandson in spelling'
2. **Seasonal teaching tools**
   'Best easter teaching tool!'
3. **Expansions and sets**
   'Great tile set! Excellent starter.'

Negative Reviews

1. **Hard instructions**
   'Instructions are complicated to follow'
2. **Incomplete kits**
   'Incomplete kit, very disappointing'
3. **Value and design**
   'Expensive for what you get'; criticism of DM screens

Summary of Key Findings

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Behaviour drives loyalty, clear customer segments, and customers are largely satisfied

1 Loyalty is Driven by Spending Behavioural

Spending score and remuneration together drive points; demographics add almost nothing

2 Reliable Model

Tuned decision tree reaches a test R^2 of 0.970

3 Five Clear Customer Segments

Each needs its own campaign, from premium rewards to low-cost nudges

4 Satisfied Customers

80 to 88% of reviews are positive; complaints focus on usability and value

What Can Turtle Games Do?

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Data-backed actions for the marketing and product teams

1 Marketing Team

- **Reward** high-income, high-spending customers with a premium tier and exclusive offers
- **Upsell** high-income, low-spending customers with personalised promotions
- **Retain** low-income, high-spending customers, who earn few points despite heavy spending
- **Promote** products as gifts and learning tools around Christmas and Easter

2 Product Team

- **Simplify** instructions and assembly guides
- **Strengthen** quality checks to stop incomplete kits
- **Redesign** Dungeon Master screens around useful reference content
- **Monitor** negative reviews (VADER below -0.05) as an early warning

Analytical Improvements

What would make the next round of analysis stronger

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1 Test alternative regression models

Try Lasso and Ridge regression for improved accuracy and feature selection

2 Use topic modelling

Identify common themes in customer reviews automatically

3 Explore other clustering methods

Try hierarchical clustering or DBSCAN for more nuanced segments, if more behavioural data were available

4 Track loyalty trends over time

Monitor loyalty and purchase frequency over time, if customer IDs and timestamps were available

Thank You

Questions and feedback are welcome

